

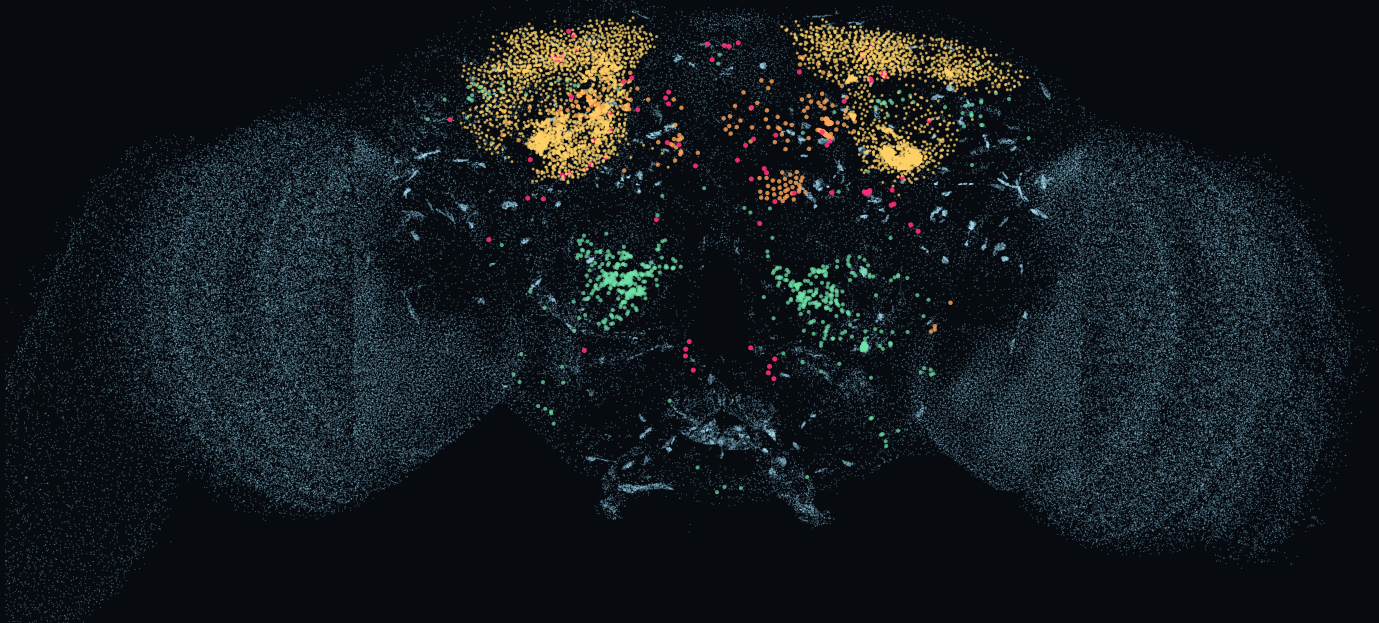
FlyScale

geometric scaling of the adult *Drosophila* connectome

139,255 neurons · 2,700,513 connections · 34,153,566 synapses · a 10x ladder

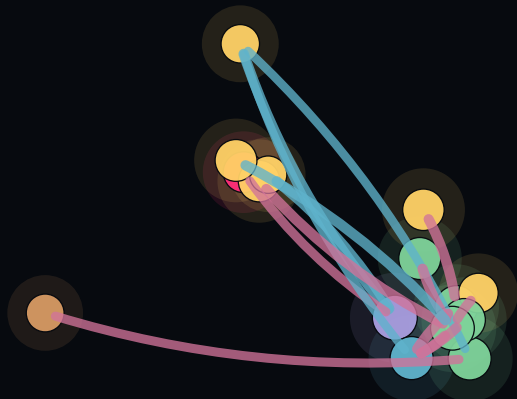
whether a connectome can be geometrically enlarged while preserving its structure, its dynamics, its capabilities and its energy budget - measured, not assumed

the whole brain - 139,255 neurons at their soma positions

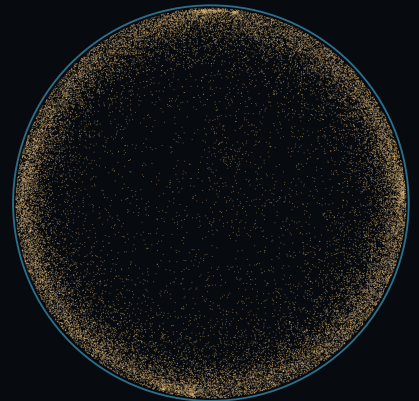


gold Kenyon cells · magenta MBON · orange DAN · green antennal-lobe projection neurons

detail: a real 15-neuron circuit (14 connections, 14303 synapses)



the fitted 2-D hyperbolic geometry



every mark on this cover is measured data. the brain is the FlyWire v783 connectome, all 139,255 neurons at their soma positions, with the mushroom-body circuit highlighted; the small panel is a real circuit (edge width = synapse count, edge colour = predicted transmitter, positions rescaled to fit); the disk is the hyperbolic geometry fitted to the same connectome in section 5.

every panel in this paper is plotted from a measured artifact - no illustrative figures